



IILM UNIVERSITY • GREATER NOIDA

An Industrial Internship Defense

1-MONTH JAVA DEVELOPER INTERNSHIP REPORT

Enterprise Backend Engineering with Spring Boot, Hibernate ORM & MySQL

PRAFFUL SHIKHAR

Roll No. 25SCS1003004010 | B.Tech CSE (AIML)

CODECTECHNOLOGIES

June 2026 – July 2026

CANDIDATE OVERVIEW

Academic Profile & Credentials



CANDIDATE NAME

PRAFFUL SHIKHAR

UNIVERSITY ROLL NUMBER

25SCSi003004010

DEGREE PROGRAMME

B.Tech Computer Science & Engineering

Specialization in AI & Machine Learning (2025–2029)



AFFILIATION

IILM UNIVERSITY

INTERNSHIP TRACK

Java Developer Track

Enterprise Backend Architecture

TRAINING DURATION

4 Weeks (01/06/2026 – 01/07/2026)

Host Organization: CodecTechnologies

An industrial software engineering consultancy delivering rigorous corporate training modeled on production Agile development lifecycles.



ENTERPRISE SYSTEMS

Production-grade Java engineering focusing on decoupled modular architectures and robust SQL databases.



AGILE SDLC MODEL

Sprint milestones, peer code reviews, Git branching practices, and automated Maven build pipelines.



HANDS-ON MENTORSHIP

Direct technical guidance in Spring Boot, Hibernate ORM persistence, and secure web interface design.

PROBLEM STATEMENT

Industry Need & Academic Gap

Academic coursework centers on standalone syntax, leaving gaps in multi-tiered enterprise architecture, ORM persistence, transactional rollback safety, and access control.



TIERED ARCHITECTURE

Decoupling UI, logic, and database access via Spring MVC.



ORM PERSISTENCE

Eliminating raw JDBC errors using Hibernate & JPA.



ROLE-BASED ACCESS

Enforcing granular permissions on sensitive employee data.



DYNAMIC UI FORMS

Constructing responsive templates via Thymeleaf.

PROJECT OBJECTIVES

Technical Milestones

ENTERPRISE JAVA MASTERY

Master OOP, Collections Framework, Streams, and Exception Handling.

SPRING MVC & BOOT

Architect scalable multi-tiered applications using Inversion of Control.

HIBERNATE ORM & JPA

Automate database persistence and manage complex entity relationships.

FULL CRUD PIPELINES

Build complete data entry, query, update, and deletion workflows.

ROLE-BASED ACCESS (RBAC)

Enforce secure data governance and multi-role user privileges.

DYNAMIC THYMELEAF UI

Render responsive client templates and automate report generation.

TECHNOLOGIES & TOOLS

The Engineering Stack



JAVA 17/21

Core Language



SPRING BOOT

Microframework



SPRING MVC

Web Controller



HIBERNATE ORM

Persistence Layer



MYSQL SERVER

RDBMS 8.0



THYMELEAF

Template UI



APACHE MAVEN

Build Automation

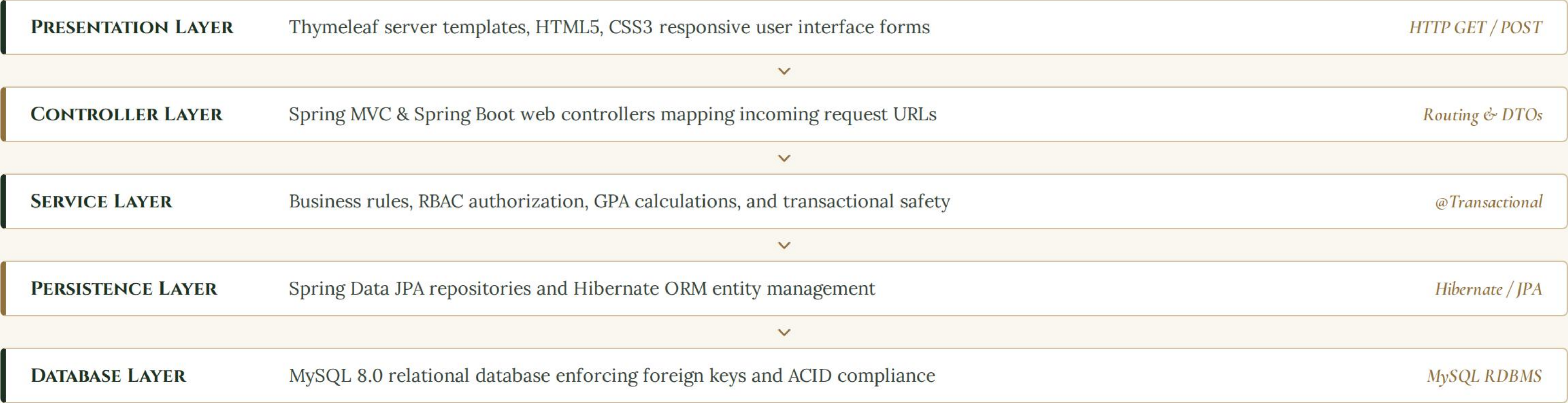


GIT & GITHUB

Version Control

SYSTEM ARCHITECTURE

Decoupled MVC Pattern



METHODOLOGY & PLAN

4-Week Execution Journey

WEEK 01

JAVA & SPRING CORE

OOP refinement, Collections, Streams, Maven setup, and Inversion of Control.

WEEK 02

EMPLOYEE SYSTEM

Spring MVC, Hibernate setup, entity mappings, and Role-Based Access Control.

WEEK 03

STUDENT SYSTEM

Spring Boot init, Spring Data JPA repositories, and GPA calculation service.

WEEK 04

UI & TESTING

Thymeleaf UI binding, form validation, integration tests, and GitHub submission.

APPLICATIONS BUILT

Primary Project Deliverables



EMPLOYEE MANAGEMENT

Spring MVC + Hibernate + MySQL

- **Full CRUD Functionality:** Automated onboarding, record updates, and profile lifecycle tracking.
- **Role-Based Access Control:** Admin and Manager permission boundaries protecting salary records.
- **Hibernate Persistence:** SessionFactory entity mapping with relational constraints.



STUDENT MANAGEMENT

Spring Boot + JPA + Thymeleaf

- **Academic Record Portal:** Student profiles, enrollment codes, and course registration.
- **Marks & GPA Computation:** Real-time grade point aggregation and performance tracking.
- **Dynamic Report Generation:** Thymeleaf template rendering for clean, printable report cards.

EXPECTED OUTCOMES

Technical Competencies Acquired



ENTERPRISE STACK COMPETENCE

Mastery in Spring Boot, Spring MVC, Spring Data JPA, and Hibernate ORM.



PRODUCTION CRUD SYSTEMS

Constructed and deployed Employee & Student Management Systems.



RELATIONAL DATA GOVERNANCE

Engineered normalized MySQL schemas with transactional ACID rollback integrity.



SDLC & TOOLING HYGIENE

Proficiency in Git branching, Maven build configurations, and project documentation.



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THANK YOU

Questions & Academic Technical Discussion

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